

30/10/25

Attention and Self-Attention

Seq 2 Seq

- Gating
 - Attention
- vanishing gradient

Attention vector $O_1 = [0.5x_1 + 0.2x_2 + 0.2x_3 + 0.1x_4]$ weighted sum of the i/p

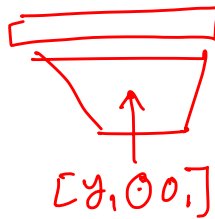
Attention distribution

0.5

0.2

0.2

0.1



unembedding layer

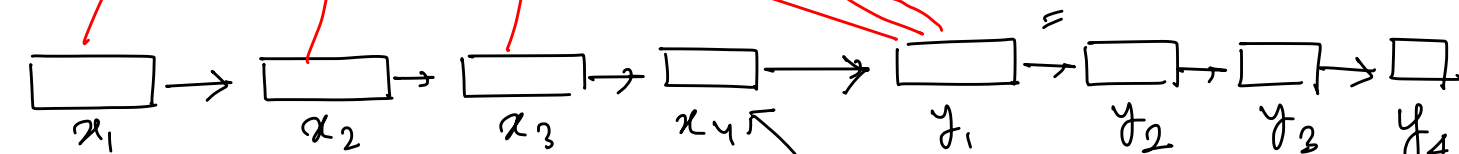
$x_1 \cdot y_1$

$x_2 \cdot y_1$

$x_3 \cdot y_1$

$x_4 \cdot y_1$

Attention score

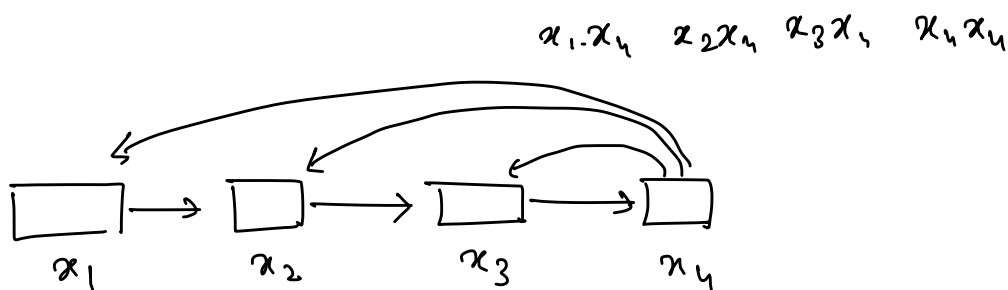


Encoder

Decoder

Conditional decoder

Bottleneck Problem



$$p_1 v_1 + p_2 v_2 + p_3 v_3 + p_4 v_4 = O_4$$

p_1

p_2

p_3

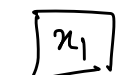
p_4

$x_1 \cdot K_1$

$x_1 \cdot K_2$

$x_1 \cdot K_3$

$x_1 \cdot K_4$



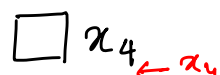
$\rightarrow K_1$
 $\rightarrow V_1$



$\rightarrow K_2$
 $\rightarrow V_2$



$\rightarrow K_3$
 $\rightarrow V_3$



$\rightarrow K_4$
 $\rightarrow V_4$
 $Q = x_4$
 $K = x_1, x_2, x_3$

	x_1	x_2	x_3	x_4
I	●			
am	///	●		
not	✓		●	///
$well$			///	///

Attention Matrix

Query

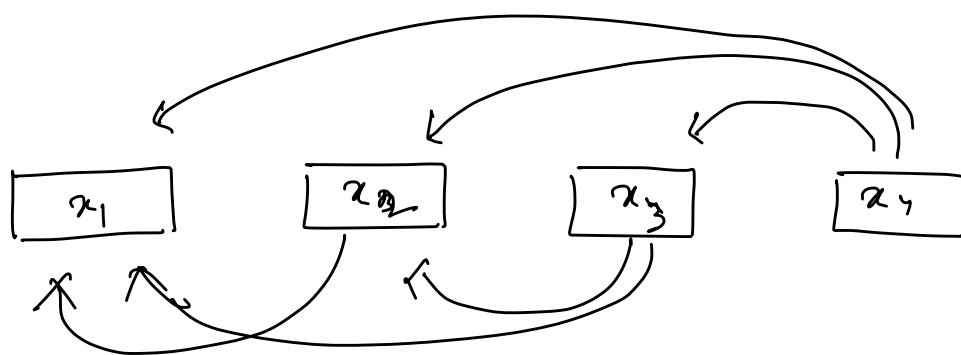
Key

Value

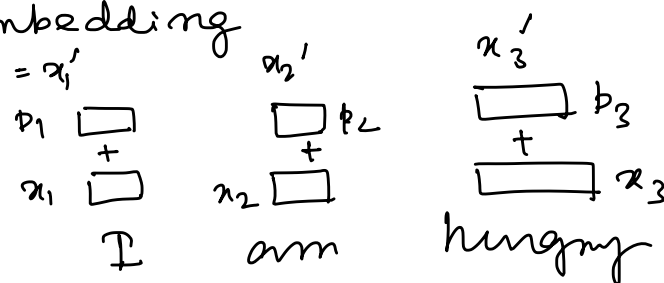
Attention vector is the weighted sum over all value vectors.

In case of normal attention, value and key vectors are the same.

Self-Attention



Positional embedding



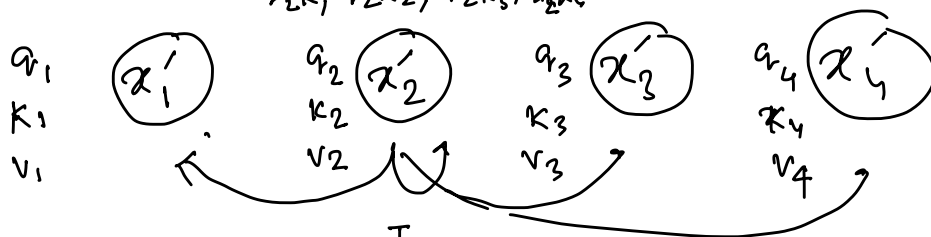
Query = W_q
Key = W_k
Value = W_v

$$O_1 = 0.7V_1 + 0.1V_2 + 0.1V_3 + 0.1V_4$$

$$\rightarrow \begin{matrix} 0.7 & 0.1 & 0.1 & 0.1 & 0.2 & 0.3 & 0.4 \end{matrix}$$

$$SM(q_1, k_1, q_1, k_2, q_1, k_3, q_1, k_4) \begin{matrix} 0.8 & 0 & 0 & 0.2 \end{matrix}$$

$$q_2, k_1, q_2, k_2, q_2, k_3, q_2, k_4$$



$$q_1 = W_q^T x_1'$$

$$q_2 = W_q^T x_2'$$

$$k_1 = W_k^T x_1'$$

$$k_2 = W_k^T x_2'$$

$$v_1 = W_v^T x_1'$$

$$v_2 = W_v^T x_2'$$

